Lab - Pushing an Image to Azure Container Registry (ACR)
To Begin with the Lab

Summary
This lab demonstrates how to push a locally built Docker image to Azure Container Registry (ACR). The process includes enabling the admin user, logging in to ACR, tagging the local image with the registry name, and pushing it to the registry so it can be pulled and run from other Azure resources.

Understanding
Azure Container Registry (ACR) is a private Docker image registry in Azure. Before pushing an image, Docker must authenticate with the registry. The local image is tagged with the ACR login server name so Docker knows the destination registry. Once pushed, the image is stored centrally and can be pulled by Azure VMs, Azure Container Instances, AKS, or other services.

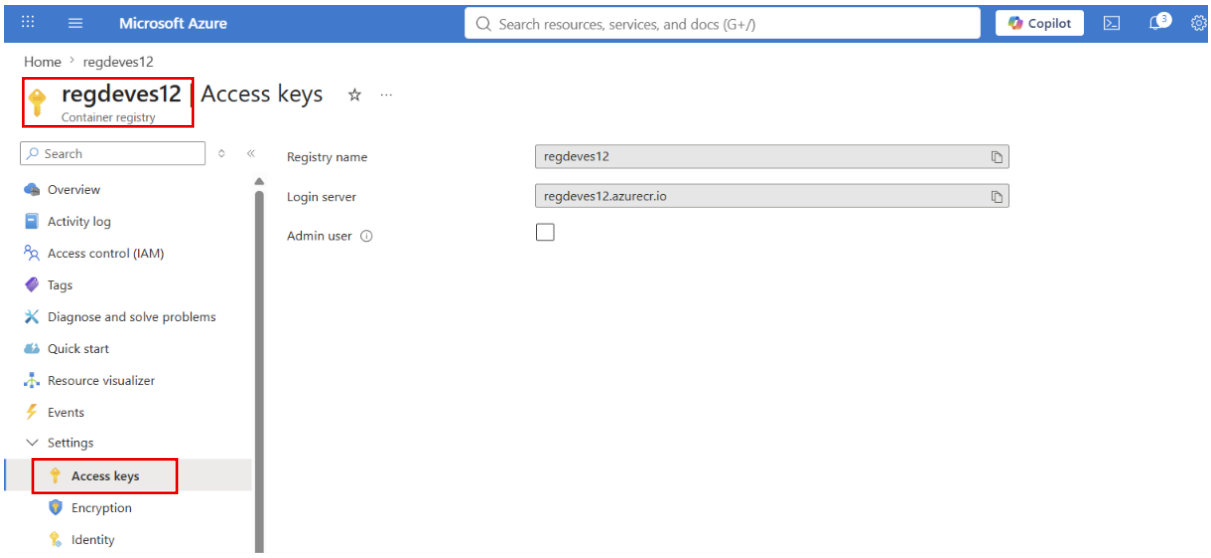
Example:

- Local Image: dockerapp:v1
- ACR Login Server: myregistry.azurecr.io
- Tagged Image: myregistry.azurecr.io/dockerapp:v1

Lab Steps

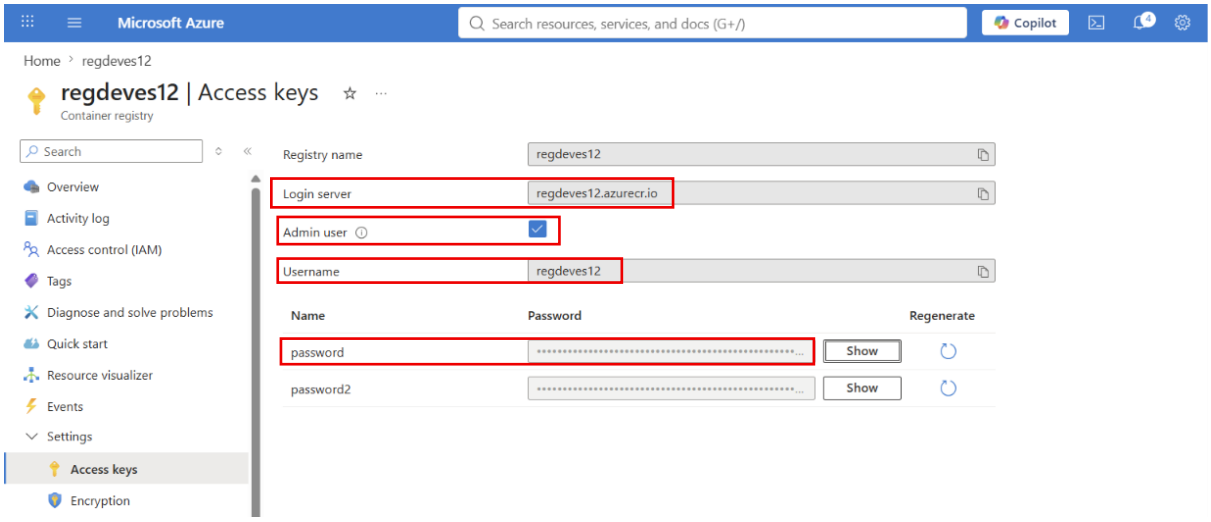
Step 1: Open Azure Container Registry

- Open the Azure Portal.
- Navigate to your **Azure Container Registry**.
- Select **Access Keys**.



Step 2: Enable Admin User

- Turn **Admin User** to **Enabled**.
- Copy the **Login Server.**
- Copy the **Username**.
- Copy either **Password** or **Password2**.

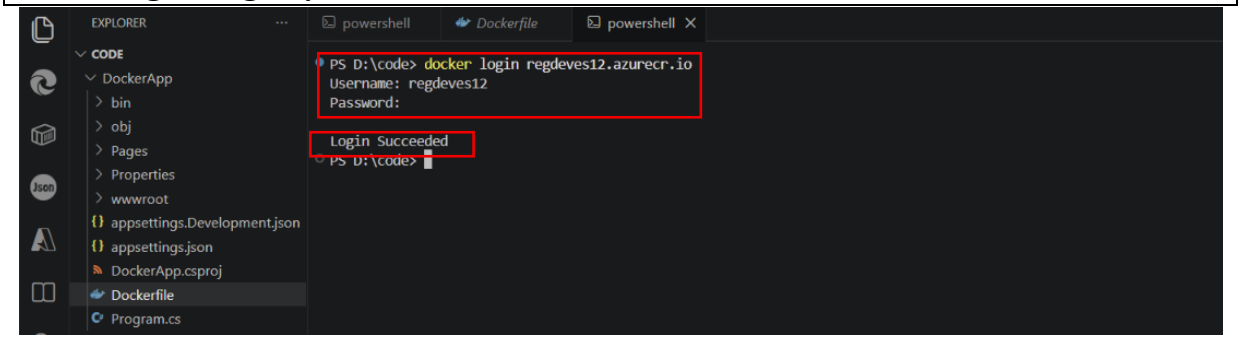


Note: In production environments, use **Azure RBAC** instead of the admin account for better security.

Step 3: Login to Azure Container Registry

- Use the ACR login server.
- Enter the **username**.
- Enter the **password**.
- **Login Succeeded.**

```
docker login <registry-name>.azurecr.io
```



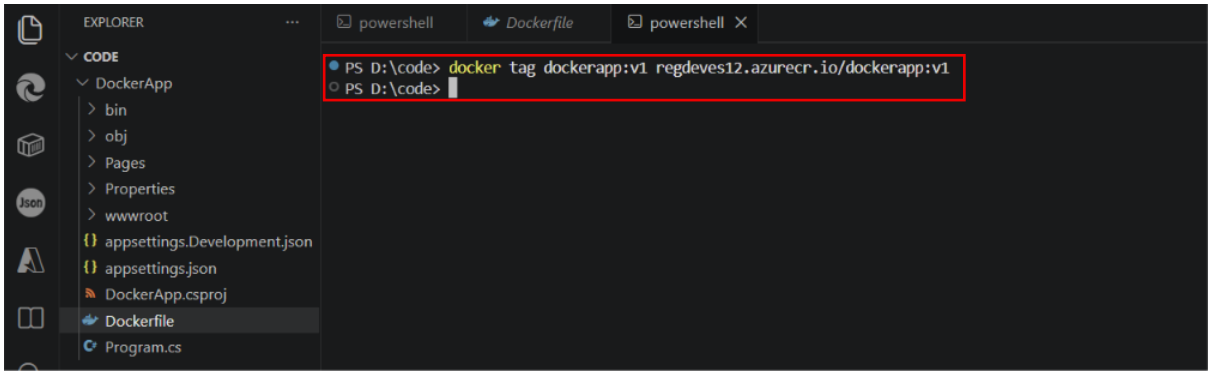
Step 4: Tag the Local Docker Image

Tag the local image with the ACR login server.

```
docker tag dockerapp:v1 <registry-name>.azurecr.io/dockerapp:v1
```

Why is tagging required?

- Docker needs to know the destination registry.
- Without the registry name, Docker assumes the image belongs to Docker Hub.
- Tagging tells Docker to push the image to Azure Container Registry.



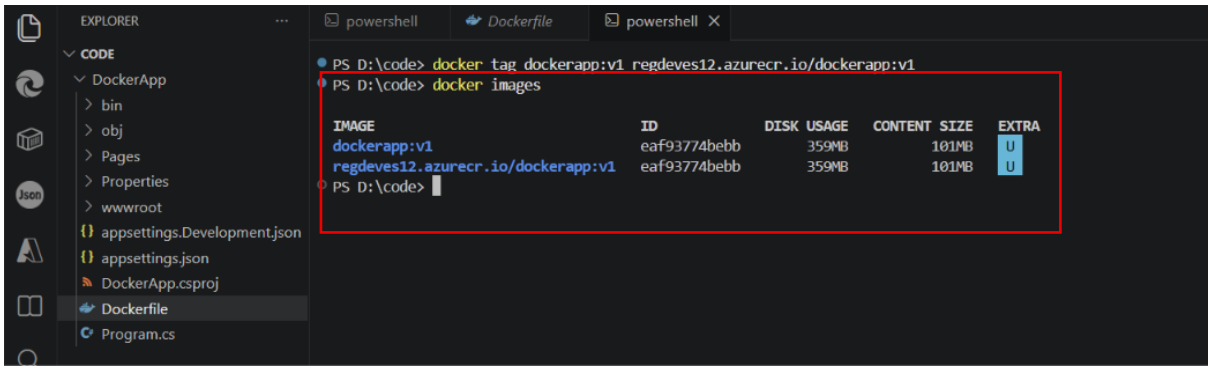
Step 5: Verify the Tagged Image

List the available Docker images.

```
docker images
```

Observe:

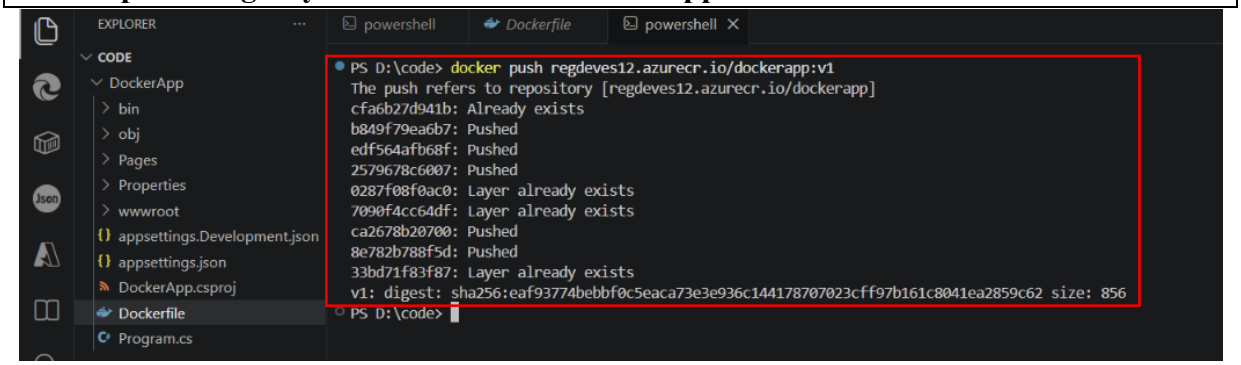
- The original image still exists.
- A new image appears with the ACR login server name.
- Both images share the same Image ID because they reference the same image.



Step 6: Push the Image to Azure Container Registry

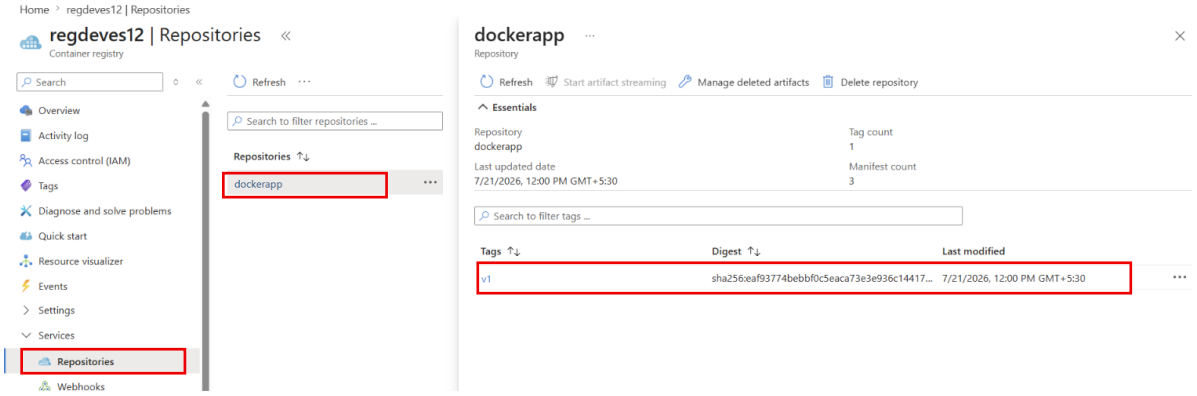
Push the tagged image.

```
docker push <registry-name>.azurecr.io/dockerapp:v1
```



Step 7: Verify in Azure Portal

- Open **Azure Container Registry**.
- Select **Repositories**.
- Confirm that the repository appears.
- Verify the image tag (v1) is listed.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1
Last updated date
7/21/2026, 12:00 PM GMT+5:30
Manifest count
3
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Tags
Digest
Last modified
v1
sha256:eaf93774bebbf0c5eaca73e3e936c14417...
7/21/2026, 12:00 PM GMT+5:30